FinanceNER: Event-based Stock Prediction from U.S. Presidential Tweets using Named Entity Recognition and Sentiment Analysis

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Abstract

As social media platforms became an increasingly relevant method of communication between a celebrity and their followers, it became more apparent that a public figure's messages to their followers can impact targeted company's revenue and reputation. This statement begged the question as to which celebrity's message could have the most impact on the stock market, and the former president Donald Trump with his affection to tweeting was chosen as the ideal candidate. We investigate the impact of U.S. president and celebrity Donald Trump's tweets from his first presidential term on financial markets through an end-to-end pipeline (FinanceNER) that combines Named Entity Recognition (NER), entity-to-ticker mapping, and sentiment analysis for event-based stock prediction. Our approach involves training a custom FinBERT-based NER model on FiNER-ORD (Shah et al., 2024), mapping the extracted entities to S&P 500 tickers, and correlating tweet sentiment with subsequent stock returns. We benchmark against a baseline, discuss the challenges of financial and political text, and investigate the predictive ability and boundaries of sentiment signals in this domain. In conclusion, tweets of the U.S. president, which has the power to influence policies of the country or influence his followers' sentiment regarding a chosen company, show some potential of being predictive of stock entities movement when paired up with sentiment analysis. The financial model of NER achieved similar accuracy to the baseline model on the annotated test data with baseline having 91% and finance NER 90%. While the accuracy is slightly lower it is important to look at other metrics and keep in mind that finance NER is trained on financial texts not tweets. Due to ambiguous multi-token entities, higher complexity of NER detection in tweets and sentiment classifications not always referring to the found entity, it is difficult to always find the right entity and make the correct prediction.

1 Introduction

Previously conducted through speeches and official press releases, political communication has changed as a result of social media platforms that allow prominent people to communicate directly with audiences around the world.

Within minutes of publication, social media, especially tweets from powerful political figures, can have a major impact on the market. We face significant challenges in quantifying this influence because financial markets and social media language are both noisy and context-dependent. The complex connections between political communications and industry-specific market responses are frequently missed by conventional sentiment analysis techniques.

Using a pipeline that combines entity-to-ticker mapping, named entity recognition (NER), and targeted sentiment analysis, our project, **FinanceNER**, tackles these issues. Financial entities were extracted from Donald Trump's presidential tweets (2017-2021) (Roy, 2021), we analyze entity-specific sentiment rather than general message tone, thus gaining a more granular understanding of potential market impacts. With this approach, we can detect when specific companies, sectors, or financial instruments are mentioned with positive or negative sentiment, providing a more precise signal for potential market movements.

We selected Trump's presidential tweets as our data source for several compelling reasons:

- their documented market-moving potential, with numerous studies showing statistically significant price movements following certain tweets

- their unprecedented frequency and directness for a sitting president
- their often explicit focus on economic and business matters
- their substantial reach, with each message potentially viewed by millions of market participants simultaneously

Through this research, we aim to answer: *How effectively can we find stock market entities using NER on presidential tweets and relate it to stock market with sentiment analysis?* Our results contribute to the broader understanding of the impact of political communication on financial markets.

2 Related Work

The relationship between textual sentiment and financial market movements has been extensively studied in multiple disciplines. Early research focused primarily on traditional news sources, with seminal work by Tetlock (Tetlock, 2007) showing that high media pessimism predicted downward pressure on market prices. As social networking became increasingly popular, researchers shifted their attention to these new information channels. Twitter-specific challenges include the prevalence of sarcasm, fleeting contextual references, and the difficulty of distinguishing between informative and noise-generating content. Sul et al. (Sul et al., 2017) noted that tweets from company executives and verified accounts carried significantly more predictive power than general Twitter sentiment, suggesting the importance of source credibility.

Recent methodological advances have attempted to address these limitations through:

- Fine-tuned language models specifically trained on financial text (FinBERT, Yang et al., 2020) (Yang et al., 2020)
- Multi-modal approaches incorporating both text and user network characteristics (Xu & Cohen, 2018) (Xu and Cohen, 2018)
- Event-based frameworks that identify discrete market-relevant announcements rather than general sentiment (Ding et al., 2019) (Ding et al., 2019)

3 Data

3.1 Tweet Corpus

We use the Complete Trump Tweets Archive (Roy, 2021), then we tokenized and labeled for NER and sentiment. Key files:

3.2 Stock Market Data

The stock market data was taken from Yahoo Finance focusing on a given ticker on the mentioned date and compared using primarily 1 day window post-tweet. Returns are calculated as percentage change in closing price.

3.3 NER Dataset and Annotation

We use FiNER-ORD from HuggingFace (Shah et al., 2024), split into train/validation. Entity labels: O (0), PER_B (1), PER_I (2), LOC_B (3), LOC_I (4), ORG_B (5), ORG_I (6).

For testing, we manually annotated 300 samples fro the Complete Trump Tweets Archive using the same criterion and labels as the training set. Due to splitting the task among 3 annotators, some ambiguous labels may not be consistently categorized across the entire set, however we tried to maintain inter-annotator agreement by consulting throughout the annotation process.

3.4 Ticker Mapping

All entities found by the NER in the test dataset other than the O labels were taken from each tweet. S&P 500 companies together with all their aliases were found on wikipedia and also new aliases were appended based of common words occurring in company names being removed, such as ”Apple Inc.” having an alias ”Apple”. The names and aliases were assigned to the corresponding ticker on the stock market. We then compared every found entity from each sentence to all company names with their aliases and saved the tickers of the companies mentioned, therefore having mapped NER detected entities to the correct firms.

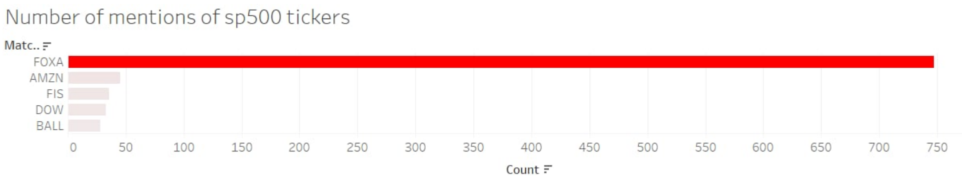

Figure 1: Number of mentions of S&P 500 tickers in Trump's tweets. FOXA is the most frequently mentioned.

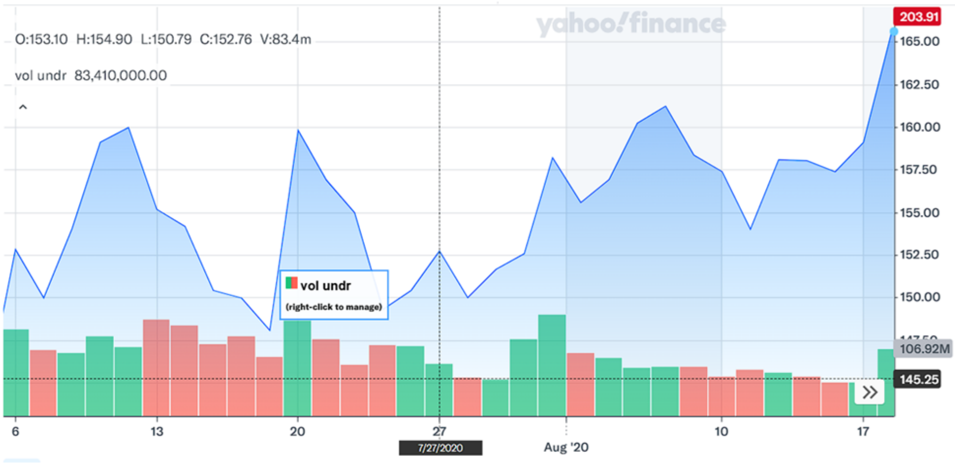

Figure 2: Example of a stock movement (AMZN) on 27th of July 2020

4 Methodology

Entity	Precision	Recall	span F1	Support
LOC_B	0.55	0.86	0.67	105
LOC_I	0.39	0.46	0.42	41
ORG_B	0.40	0.14	0.20	162
ORG_I	0.43	0.11	0.18	170
PER_B	0.78	0.48	0.60	248
PER_I	0.72	0.10	0.18	390
micro avg	0.60	0.28	0.38	1128
macro avg	0.30	0.20	0.21	1128
weighted avg	0.61	0.28	0.33	1128

Table 1: Baseline NER performance on the annotated test set.

4.1 Baseline Model

The baseline uses a simple sequence labeling approach (e.g., CRF or logistic regression) and is evaluated in using the same tests as our transformer-based model for comparison. Trained using the EWT portion of the Universal NER project, the baseline is shown to perform better in predicting PER_B classes resulting in higher overall accuracy of 91.45%. However, this can be largely attributed to the skewed distribution of entities in our test set, which unlike the training set and typical financial texts, are not in favor of organization entities. Due to the larger size of the EWT dataset, the model was more likely to come across person entity labels and accurately categorize them. (Andrew and Gao, 2007).

4.2 Custom NER Model Hyperparameters

Our NER model is based on FinBERT, a BERT variant for finance, fine tuned to account for noise and limited training size We use `BertForTokenClassification` with 12 layers, 12 attention heads, hidden size 768. The tokenizer is adapted for tweets: lowercasing, special tokens, truncation at 128 tokens. Training uses:

- **Learning rate (2e-5):** A small learning rate helps avoid dramatic change in weights of FinBERT's pretrained financial language understanding while allowing gradual domain adaptation.

- **Batch size (32):** Used widely in many hugging face and research paper examples, we chose an average small batch size to improve convergence and achieve more stable gradient descent. While this might overfit slightly on noise / shorter training samples, our training set consists entirely on high quality annotations of longer financial articles.

- **Epochs (2):** Due to the similarity in domain between FinBert's pretraining and our Finer-ord training set, we anticipated that it would not be necessary to run many epochs and risk overfitting. We observed that beyong 2 epochs, the training loss would not improve.

- **Optimizer (AdamW):** AdamW is the standard for transformer fine-tuning, helping with generalization by decoupling weight decay from gradient updates.
- **Loss (Cross-entropy):** Very compatible with gradient descent and suitable for multi-label classification.

4.3 Pipeline Overview

1. **NER Module:** Detects/classifies financial entities in tweets.
2. **Entity-Ticker Mapping:** Links NER output to S&P 500 tickers.
3. **Sentiment Analysis:** Assigns sentiment (positive, negative, neutral) to NER-filtered tweets.
4. **Stock Movement Correlator:** Manually linking sentiment-labeled tweets to stock returns over 1d windows.

5 Results

5.1 NER Evaluation

NER is evaluated on using precision, recall, and span F1 for each entity type. The overall accuracy is 90.67%.

Entity	Precision	Recall	span F1	Support
LOC_B	0.53	0.87	0.66	15
LOC_I	0.29	0.61	0.40	41
ORG_B	0.30	0.37	0.33	27
ORG_I	0.36	0.09	0.15	53
PER_B	0.78	0.37	0.50	24
PER_I	0.71	0.11	0.19	39
micro avg	0.48	0.29	0.36	1128
macro avg	0.27	0.22	0.20	1128
weighted avg	0.57	0.29	0.32	1128

Table 2: NER performance on the annotated test set.

5.2 NER & stock entities detection

The entity mapping to the stock market listed companies worked successfully. Figure 1 shows the distribution of S&P 500 ticker mentions in the tweet corpus. FOXA stands out as the most frequently mentioned ticker, reflecting the president's focus on Fox Corporation in his tweets.

5.3 Sentiment and Stock Correlation

We analyze the correlation between tweet sentiment and stock returns over 1 day window. We tested different chosen entities for stock movement based of the prediction. While some positive sentiment tweets precede upward price moves, overall correlation is modest, confirming prior literature (Ando and Zhang, 2005). Further work would require automatizing this process, testing different time windows for the prediction, and checking quantified correlation for a broader set of tickers.

5.4 Example showcase: Stock reaction to Tweeted entities

To illustrate an application of the **FinanceNER** pipeline, we examined a notable tweet from July 27, 2020, where President Trump stated: ”The Washington Post is a political front for Amazon. Nobody treated Ronald Reagan worse!” This case exemplifies the complexity of entity detection in political discourse, particularly when entities are referenced indirectly. In this instance, Amazon was mentioned through its ownership relationship with The Washington Post rather than through direct reference.

Our NER model successfully identified ”Amazon” as an ORG entity despite this indirect reference pattern and correctly mapped it to the AMZN ticker. The sentiment analysis component classified this statement as conveying negative sentiment toward the entity. As shown in Figure 2, AMZN experienced a notable price movement in the subsequent trading session, with decreased trading volume accompanying the price change. This pattern aligns with our working hypothesis regarding the potential correlation between presidential tweet sentiment and short-term market reactions, though causality cannot be definitively established given the multitude of factors that influence stock prices.

6 Analysis

6.1 Error Analysis

In sentiment analysis, sarcasm and political satire—frequent in Trump's tweets—pose challenges. For instance ”so great that oil prices are falling, (, thank you president t)” is framed with a positive sentiment, however while this may be good news for trump, this reflect negative price progression resulting in contradictory predictions for tagged entities.

NER errors are most common for ambiguous or multi-token entities. Sentiment misclassifications occur in sarcastic or indirect tweets. Not all tweets mentioning financial entities are market-moving; external events and broader context play a major role. Stock price movements are influenced by many factors beyond tweet sentiment. Other mislabeling may be cause by domain-specific model (trained on formal, structured, financial texts) to slightly out-of-domain data, where unseen or rare labels used in ambiguous context create label confusion.

The most frequent classes of errors:

- **Multi-token entities:** Entities such as ”Pensacola Florida” (’pens’, ’aco’, ’la’, ’,’, ’florida’) sometimes only have the first (few) tokens tagged.
- **Entity label confusion:** PER_I tokens were often misclassified due to annotation inconsistencies or insufficient training examples in the FiNER-ORD dataset. Additionally, the model often confuses ORG_I with LOC_I due to contextual overlap (e.g., “New York” used in both senses).
- **Financial entity types bias:** Some terms in political discourse may not exist in financial texts, which combined with the modle potentially overfitting to financial context can caused some PER and LOC entities to be labelled as ORG, and other unseen ORG entities, such as ”fda” to be ignored.
- **Misalignment in informal language:** Twitter-specific formatting for accounts representing PER or ORG are not consistently well-handled by the tokeniser used for Finer-ORD from lack of input representation

6.2 Limitations

Our dataset is limited to one politician and a fixed time period. The NER model struggles with ambiguous and multi-token entities, and sentiment analysis is challenged by the rhetorical complexity of political tweets. For instance place names such as US states can refer to LOC, PER, or ORG depending on context. Due to resource limitations, our test data size may not be a representative sample of the overall Tweets archive, making it difficult to detect trends in errors for financial entity detection due to their scarcity. Stock price movements are also affected by a multitude of factors and more layers can be added to a simple NER model to account for other features and metadata.

7 Conclusion

Finance NER scored a similar accuracy of 90% compared to the baseline NER of 91%. The slight difference can be attributed to the fact that tweets are not conventional financial articles and are a lot more ambiguous when it comes to entities. It is also important to keep in mind other metrics than accuracy when comparing the two result tables above. Finance NER provides a reproducible pipeline for event-based stock prediction from political tweets, combining custom financial NER, robust entity-ticker mapping, and sentiment analysis. While sentiment alone is a weak predictor, accurate NER and mapping are essential for any study in this domain, our results highlight the challenges of sentiment-based prediction and the need for further integration of contextual and market data.

8 Future Work

- Incorporate more advanced sentiment models (e.g., transformer models fine-tuned on political text).
- Expand the dataset to include other politicians and recent tweets.
- Integrate event detection and market context for improved modeling.
- Automate the stock movement correlation calculations based on the sentiment analysis and test more quantitative measures of comparison.

Appendix

GitHub Repository

8.1 Chatbot usage disclosure :

In line with the ACL 2023 policy, we disclose that we mainly used chatbots for literature search assistance at the start of the project and as an assistance for parts of the code. Later on we only used ChatGPT for questions regarding LaTeX syntax, researching methods for building more efficient NER training pipelines.

8.2 Group Contribution:

We split tasks among group members but worked together on most topics. Manual annotation was divided evenly.

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